
Funktionentheorie

Autumn Semester 2026

Lecture Notes

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References

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Funktionentheorie

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1 Introduction

Lec 1

1.1 Motivation

Consider the following example. On $\mathbb{R}$ it is not too difficult to construct a function f that is differentiable n times but not differentiable $n+1$ times. For example,

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x^2}) & x \neq 0, \\ 0 & x = 0 \end{cases}.$$

This function is differentiable but the derivative is not continuous. Integrating, we find a function that is differentiable twice but not three times. In fact, we can construct a function that is differentiable n times but not $n+1$ times for any $n \in \mathbb{N}$ using this method.

In contrast, we'll see that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is differentiable once, then it is differentiable infinitely many times.

Another example is that for real numbers, there are functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which are infinitely differentiable but not analytic. I.e. the Taylor series at some point x_0 exists, but does not converge to the function in any neighborhood of x_0 . For example,

$$x \mapsto (x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0, \\ 0 & x = 0 \end{cases},$$

where all derivatives are zero so the Taylor series is identically zero.

In contrast, we'll see that a differentiable function $f : \mathbb{C} \rightarrow \mathbb{C}$ has a power series representation. So differentiability is equivalent to analyticity.

A further comparison is that there are many examples of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which are C^∞ and bounded, for example

$$f(x) = \sin(x).$$

In contrast, we'll see that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is bounded and differentiable, then it is constant.

The robustness of this theory can be used for applications which seem to have nothing to do with complex analysis. Probably the most used examples are evaluating real integrals.

$$\int_0^\infty \cos(t^2) dt = \int_0^\infty \sin(t^2) dt = \frac{\sqrt{2\pi}}{4}.$$

Another famous example is the prime number theorem, which states that the number of primes less than x is asymptotic to $\frac{x}{\log(x)}$.

Another classic is to consider a non-zero polynomial $f(x) \in \mathbb{C}[x]$. Then the Fundamental Theorem of Algebra states that f has a zero in $\mathbb{C}$.

Lastly, consider

$$r_4(n) = \#\{(x, y, z, w) \in \mathbb{Z}^4 : x^2 + y^2 + z^2 + w^2 = n\}.$$

Lagrange proved that every positive integer has such a representation, namely $r_4(n) \geq 1$. Jacobi later proved that

$$r_4(n) = 8 \sum_{d|n; 4 \nmid d} d.$$

1.2 Review of Complex Numbers

Recall that the complex numbers are given as

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\},$$

where $i^2 = -1$. We can embed $\mathbb{R}$ into $\mathbb{C}$ via $x \mapsto x + i0$. We also defined

Definition 1.1: Real and Imaginary Parts

For $z = x + iy \in \mathbb{C}$, we define the **REAL PART** $\Re(z) = x$ and the **IMAGINARY PART** $\Im(z) = y$.

Furthermore, the **COMPLEX CONJUGATE** is defined as

$$\bar{z} = x - iy.$$

From this it follows that

Lemma 1.2:

For $z = x + iy \in \mathbb{C}$, we have

$$z \in \mathbb{R} \Leftrightarrow z = \bar{z} \quad z \in i\mathbb{R} \Leftrightarrow z = -\bar{z}.$$

$\mathbb{C}$ can also be represented by pairs of real numbers in $\mathbb{R}^2$. For example $z = (x, y), w = (u, v) \in \mathbb{R}^2$ where

$$z = w \Leftrightarrow x = u, y = v.$$

Addition can be defined in both notations as

$$z + w = (x + u, y + v) \quad z + w = (x + u) + i(y + v).$$

Multiplication was a bit more cumbersome. Using the definition of i , we have

$$\begin{aligned} z \cdot w &= (x + iy)(u + iv) \\ &= xu + i^2 yv + i(xv + yu) \\ &= (xu - yv) + i(xv + yu). \end{aligned}$$

So in the pair representation, multiplication is given by

$$(x, y) \cdot (u, v) = (xu - yv, xv + yu).$$

Note that in the pair representation,

$$i \leftrightarrow (0, 1) \quad \text{and} \quad 1 \leftrightarrow (1, 0).$$

The good thing about the pair notation, $(\mathbb{R}^2, +, \cdot)$ satisfies

1. $(\mathbb{R}^2, +)$ is an abelian group with identity $(0, 0)$
2. $(\mathbb{R}^2 \setminus \{(0, 0)\}, \cdot)$ is an abelian group with identity $(1, 0)$
3. The distributive law holds: $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$

Together, these properties make $(\mathbb{R}^2, +, \cdot)$ a field. So the complex numbers with the above addition and multiplication form a field.

The additive inverse of $z = (x, y)$ is given by $-z = (-x, -y)$. The multiplicative inverse of $0 \neq z = (x, y)$ is given by

$$z^{-1} = \frac{\bar{z}}{|z|^2} = \frac{(x, -y)}{x^2 + y^2} = \left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right).$$

Here, $|z| \in \mathbb{R}$ is the **MODULUS** or **NORM** of z , defined as

$$|z| = \sqrt{x^2 + y^2}.$$

We can also consider the **POLAR COORDINATES** of z . Any point on the plane $\mathbb{R}^2$ can be represented by a distance r from the origin and an angle θ from the positive x -axis. Here,

$$z = r(\cos(\theta) + i\sin(\theta)) = re^{i\theta}.$$

Beware that the representation is not unique, since θ can be replaced by $\theta + 2\pi k$ for any $k \in \mathbb{Z}$, unless we restrict θ to a certain interval. The standard is to take $\theta \in (-\pi, \pi]$. θ is called the **ARGUMENT** of z , denoted $\arg(z)$, and r is the modulus $|z|$. For the argument in $(-\pi, \pi]$, we write $\text{Arg}(z)$ and call it **PRINCIPAL ARGUMENT**.

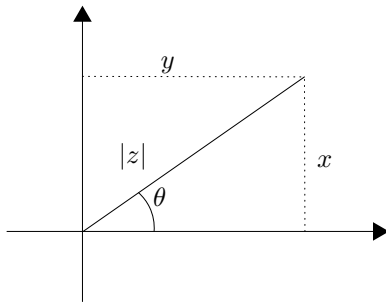


Figure 1: Polar coordinates of a complex number z

Lec 2

The principal argument of $z = x + iy \neq 0$ we have

$$\text{Arg}(z) = \begin{cases} \arcsin\left(\frac{y}{|z|}\right) & x \geq 0 \\ \pi - \arcsin\left(\frac{y}{|z|}\right) & x < 0, y \geq 0 \\ -\pi - \arcsin\left(\frac{y}{|z|}\right) & x < 0, y < 0 \end{cases}.$$

Here, for $t \in [-1, 1]$, $\arcsin(t)$ is the unique number u in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $\sin(u) = t$.

Notice, that

$$\arg(z^{-1}) = -\arg(z) \quad \text{and} \quad \arg(zw) = \arg(z) + \arg(w).$$

But, it is not always the case if we use the principal argument instead. For example

$$\text{Arg}\left(-\frac{1}{2}\right) = \pi \neq -\text{Arg}(-2) = -\pi.$$

Lemma 1.3:

Let $z, w \in \mathbb{C}$. Then

- $|z| = 0 \Leftrightarrow z = 0$
- $||z| - |w|| \leq |z + w| \leq |z| + |w|$
- $|zw| = |z||w|$
- $|\bar{z}| = |z|$
- $|\Re(z)| \leq |z|$ and $|\Im(z)| \leq |z|$

If we have two numbers $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ in polar form, then

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

$\mathbb{C}$ also has a representation in terms of 2×2 matrices. For $z = a + ib$, let

$$Z = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

If $w = c + id$, then

$$ZW = \begin{pmatrix} ac - bd & -(bc + ad) \\ bc + ad & ac - bd \end{pmatrix}.$$

So we can do the multiplication of complex numbers using matrix multiplication.

We can also write the matrix representation as

$$Z = \Re(z) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \Im(z) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

In polar form, $z = re^{i\theta}$, we have

$$Z = \begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix} \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

Notice that the second matrix lives in $SO(2)$, the group of rotations in $\mathbb{R}^2$.

1.3 Notations from Analysis

We like to recall some definitions from analysis regarding open and closed sets.

Definition 1.4: Discs

The **OPEN DISC** of radius r centered at z is the set

$$D_r(z) = \{w \in \mathbb{C} : |w - z| < r\}.$$

The **CLOSED DISC** of radius r centered at z is the set

$$\bar{D}_r(z) = \{w \in \mathbb{C} : |w - z| \leq r\}.$$

The boundary circle is the set

$$C_r(z) = \{w \in \mathbb{C} : |w - z| = r\}.$$

Definition 1.5: Connected and Disconnected Sets

A subset $A \subset \mathbb{C}$ is called **DISCONNECTED** if there exists two open sets $U, V \subset \mathbb{C}$ such that

$$U \cap V = \emptyset, \quad A \cap U \neq \emptyset, \quad A \cap V \neq \emptyset, \quad A \subset U \cup V.$$

A set is called **CONNECTED** if it is not disconnected. A connected open subset $U \neq \emptyset$ of $\mathbb{C}$ is called a **REGION** or a **DOMAIN**.

1.4 Holomorphic Functions

We now want to define what it means for a complex function to be differentiable.

Definition 1.6: Holomorphic functions

Let $\Omega \subset \mathbb{C}$ be an open and $f : \Omega \rightarrow \mathbb{C}$ a function.

f is called **DIFFERENTIABLE** or **HOLOMORPHIC** at $z_0 \in \Omega$ if

$$\lim_{z \rightarrow z_0, z \neq z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{h \rightarrow 0, h \neq 0} \frac{f(z_0 + h) - f(z_0)}{h}.$$

If f is differentiable $\forall z \in \Omega$, it is holomorphic on Ω . If $\Omega = \mathbb{C}$ and f is holomorphic on $\mathbb{C}$ then f is called **ENTIRE**.

The value of the limit is denoted by $f'(z_0)$.

Example 1.7:

Consider $f : \mathbb{C} \rightarrow \mathbb{C}, z \mapsto z$. Then f is entire. Indeed,

$$\lim_{h \rightarrow 0, h \neq 0} \frac{f(z_0 + h) - f(z_0)}{h} = \lim_{h \rightarrow 0, h \neq 0} \frac{z_0 + h - z_0}{h} = 1.$$

Proposition 1.8: Properties of holomorphic functions

1. Let $\mathcal{H}(\Omega)$ be the set of holomorphic functions on Ω . Then $\mathcal{H}(\Omega)$ is a vector space over $\mathbb{C}$.

Here, $0 : z \mapsto 0$ is the 0-element.

2. If $f, g \in \mathcal{H}(\Omega)$ then so is $f \cdot g$. Also, we have

$$(fg)'(z_0) = f'(z_0)g(z_0) + g'(z_0)f(z_0).$$

3. Lastly if $g(z_0) \neq 0$, for any $z_0 \in \Omega$, then f/g is holomorphic at z_0 and

$$\left(\frac{f}{g}\right)'(z_0) = \frac{f'(z_0)g(z_0) - g'(z_0)f(z_0)}{g(z_0)^2}.$$

4. If $f : \Omega \rightarrow U$ and $g : U \rightarrow \mathbb{C}$ are holomorphic, then $g \circ f$ is holomorphic and

$$(g \circ f)'(z_0) = g'(f(z_0))f'(z_0).$$

Proof. The proof is the same as in real analysis. We will thus only show the proof of the last statement.

Let $z_0 \in \Omega$ and let $f(z_0) = w_0 \in U$. Consider

$$F : \Omega \rightarrow \mathbb{C}, G : U \rightarrow \mathbb{C}.$$

where

$$F(z) = \begin{cases} \frac{f(z) - f(z_0)}{z - z_0} & z \neq z_0 \\ f'(z_0) & z = z_0 \end{cases}, \quad G(w) = \begin{cases} \frac{g(w) - g(w_0)}{w - w_0} & w \neq w_0 \\ g'(w_0) & w = w_0 \end{cases}.$$

Since f and g are holomorphic, F and G are continuous. Hence, $G \circ f$ is continuous at z_0 .¹ Now,

$$\frac{(g \circ f)(z) - (g \circ f)(z_0)}{z - z_0} = \frac{g(f(z)) - g(f(z_0))}{f(z) - f(z_0)}.$$

Using $f(z_0) = w_0$ we can write this as

$$\frac{g(f(z)) - g(w_0)}{f(z) - w_0} \cdot \frac{f(z) - f(z_0)}{z - z_0} = G(f(z))F(z).$$

Note that if $f(z) = w_0$ we get that $F(z) = \frac{w_0 - f(z_0)}{z - z_0} = 0$. Hence, taking the limit $z \rightarrow z_0$ and using continuity, we get that

$$\begin{aligned} \lim_{z \rightarrow z_0, z \neq z_0} \frac{(g \circ f)(z) - (g \circ f)(z_0)}{z - z_0} &= \lim_{z \rightarrow z_0, z \neq z_0} G(f(z))F(z) \\ &= G(f(z_0))F(z_0) \\ &= g'(f(z_0))f'(z_0). \end{aligned}$$

□

Remark

If $f : \Omega \rightarrow \mathbb{C}$ is differentiable at $z_0 \in \Omega$, then $\exists$ a complex number $c \in \mathbb{C}$ such that

$$f(z) = f(z_0) + c(z - z_0) + o(|z - z_0|) \quad \text{as } z \rightarrow z_0.$$

Here, for c we write $f'(z_0)$.

Example 1.9: Polynomials are entire

Using $f(z) = z$ entire and Proposition 1.8, we get that any polynomial is entire. If $g(z) = z^n$, then

$$g'(z) = nz^{n-1}.$$

Example 1.10: Real and complex differentiability

Let $f(z) = \bar{z}$. Then

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\overline{z_0 + h} - \bar{z}_0}{h} = \frac{\bar{h}}{h}.$$

If we choose $h = t, t \in \mathbb{R}$, then $\frac{\bar{h}}{h} = 1$. If we choose $h = it, t \in \mathbb{R}$, then $\frac{\bar{h}}{h} = -1$. Hence, the limit does not exist and f is not holomorphic.

If we look at f as a function from $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, then $\tilde{f}(x, y) = (x, -y)$. Then $\tilde{f}$ is differentiable in the real sense and the Jacobian is given by

$$D\tilde{f}(x, y) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Recall, that a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, (x, y) \mapsto (u, v)$ is differentiable at (x_0, y_0) if there exists a linear map $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that

$$\lim_{(x, y) \rightarrow (x_0, y_0)} \frac{|f(x, y) - f(x_0, y_0) - L(x - x_0, y - y_0)|}{|(x - x_0, y - y_0)|} = 0.$$

The linear map L is unique and called the **DIFFERENTIAL** Df . In the standard basis, the linear map L has a matrix representation which is called the **JACOBIAN** of f . It is given by

$$Jf(x_0, y_0) = \begin{pmatrix} \frac{\partial u}{\partial x}(x_0, y_0) & \frac{\partial u}{\partial y}(x_0, y_0) \\ \frac{\partial v}{\partial x}(x_0, y_0) & \frac{\partial v}{\partial y}(x_0, y_0) \end{pmatrix}.$$

¹ f is differentiable at z_0 so continuous at z_0 .

Recall from linear algebra, that we can view $\mathbb{C}$ as a vector space over $\mathbb{C}$ with dimension 1 with basis $\{1\}$. Alternatively we can view $\mathbb{C}$ as a vector space over $\mathbb{R}$ with dimension 2 with basis $\{1, i\}$.

A map $T : \mathbb{C} \rightarrow \mathbb{C}$ is $\mathbb{C}$ -linear if $Tz = \lambda z$ for some $\lambda \in \mathbb{C}$. It is $\mathbb{R}$ -linear if

$$T(x + iy) = xT(1) + yT(i) \quad \forall x, y \in \mathbb{R}.$$

Using $x = \frac{z+\bar{z}}{2}$ and $y = \frac{z-\bar{z}}{2i}$, we can write

$$T(z) = \underbrace{\frac{T(1) - iT(i)}{2}}_{:=\lambda} z + \underbrace{\frac{T(1) + iT(i)}{2}}_{:=\mu} \bar{z}.$$

Since T is linear, the coefficient of $\bar{z}$ must be 0. So every $\mathbb{C}$ -linear map is $\mathbb{R}$ -linear if we set $\mu = 0$. However an $\mathbb{R}$ -linear map is only $\mathbb{C}$ -linear if $\mu = 0$, so

$$T(i) = iT(1).$$

If $T(1) = a + ib$ and $T(i) = c + id$, then the condition $T(i) = iT(1)$ is equivalent to

$$a = d \quad \text{and} \quad b = -c.$$

Every $\mathbb{R}$ -linear map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by a 2×2 matrix $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$. If this $\mathbb{R}$ -linear map is also $\mathbb{C}$ -linear, then on the matrix we have $a = d$ and $b = -c$. Hence, the matrix is of the form

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

Lec 3

1.5 Cauchy-Riemann equations

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a holomorphic function at $z = z_0$. If we write $f(x + iy) = u(x, y) + iv(x, y)$. If we view f as a map from $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, if f is differentiable at z_0 , then

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

The limit exists and is independent of the way we approach z_0 . In particular, we can approach z_0 along the line $z = x + iy_0$ and let $x \rightarrow x_0$. Thus,

$$\begin{aligned} f'(z_0) &= \lim_{x \rightarrow x_0} \frac{f(x + iy_0) - f(x_0 + iy_0)}{x - x_0} \\ &= \lim_{x \rightarrow x_0} \frac{u(x - x_0) - u(x_0, y_0)}{x - x_0} + i \lim_{x \rightarrow x_0} \frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} \\ &= u_x(z_0) + iv_x(z_0). \end{aligned}$$

We can also approach z_0 along the line $z = x_0 + iy$ and let $y \rightarrow y_0$. Doing the same calculation, we find

$$f'(z_0) = v_y(z_0) - iu_y(z_0).$$

Combining the equations, we find the **CAUCHY-RIEMANN EQUATIONS**

Theorem 1.11: Cauchy-Riemann equations

For a function $f : \mathbb{C} \rightarrow \mathbb{C}$, $f(z) = u(z) + v(z)$, if f is holomorphic at z_0 , then

$$u_x(z_0) = v_y(z_0), \quad v_x(z_0) = -u_y(z_0).$$

Definition 1.12:

We define the operator

$$\frac{\partial}{\partial z} := \frac{1}{z} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right)$$

and

$$\frac{\partial}{\partial \bar{z}} := \frac{1}{z} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$$

Proposition 1.13:

If f is holomorphic at z_0 , $f(z) = u + iv$, then

$$\frac{\partial f}{\partial \bar{z}} = 0,$$

and

$$f'(z_0) = \frac{\partial}{\partial z} f(z_0) = 2 \frac{\partial u}{\partial z}(z_0).$$

If we write $f(z) = \tilde{f}(x, y)$ as a function from $\mathbb{R}^2$ to $\mathbb{R}^2$, then $\tilde{f}$ is differentiable with Jacobian

$$J_{\tilde{f}}(x_0, y_0) = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} u_x & u_y \\ -u_y & u_x \end{pmatrix}.$$

Furthermore, $\det J_f = |f'(z_0)|^2 = u_x^2 + u_y^2 = u_x^2 + v_x^2$

Proof. Per definition

$$\begin{aligned} \frac{\partial f}{\partial \bar{z}} &= \frac{1}{z} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) (u + iv) \\ &= \frac{1}{2} (u_x + iv_x + i(u_y + iv_y)) \\ &= \frac{1}{2} (u_x - v_y + i(v_x + u_y)) = 0, \end{aligned}$$

where we used the Cauchy Riemann equations in the last step.

Using the equations we derived for the CR-Equations we get

$$f'(z_0) = \frac{1}{2} (u_x + iv_x + v_y - iu_y) = \frac{1}{2} (f_x(z_0) - if_y(z_0)).$$

But this is exactly $\frac{\partial f}{\partial z} f(z_0)$

The CR equation give

$$f'(z_0) = \frac{\partial}{\partial z} f(z_0) = u_x + iv_x = u_x - iu_y.$$

Reformulating, we find the result.

If $z_0 = x_0 + iy_0$ and $h = h_1 + h_2 i$, then since f is holomorphic at z_0 ,

$$f(z_0 + h) = f(z_0) + f'(z_0)h + h\varepsilon(h),$$

where $\lim_{h \rightarrow 0} \varepsilon(h) = 0$.

If $f'(z_0) = a + ib$, where $f' = u_x + iv_x = v_y - iu_y$, then

$$\begin{aligned} f'(z_0)h &= (a + ib)(h_1 + h_2 i) \\ &= (ah_1 - bh_2) + i(bh_1 + ah_2). \end{aligned}$$

If we write $\tilde{f}(x, y) = f(z)$ and $H = (h_1, h_2)$, then

$$\left[\tilde{f}((x_0, y_0) + (h_1, h_2)) - \tilde{f}(x_0, y_0) - \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \right] / |H|,$$

goes to zero.

This says, that $\tilde{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is differentiable with

$$J_{\tilde{f}}(x_0, y_0) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} u_x & -v_x \\ v_x & u_x \end{pmatrix}.$$

Hence, $\det J_{\tilde{f}} = a^2 + b^2 = |f'(z_0)|^2 = u_x^2 + v_x^2 = u_x^2 + u_y^2$.

□

Lec 4

Note, that the first equation,

$$\frac{\partial f}{\partial \bar{z}} = 0,$$

is a compact way to write the two Cauchy-Riemann equations.

Theorem 1.14:

Suppose $f = u + iv$ is a function $f : \Omega \rightarrow \mathbb{C}$. If u and v are continuously differentiable and satisfy the Cauchy-Riemann equations on Ω , then f is holomorphic.

Note, that it would only be necessary to have that u, v are totally differentiable.

Proof. Let $z_0 = (x_0, y_0) \in \Omega$ and $h = (h_1, h_2) \in \mathbb{C}$. Since u, v are continuously differentiable, we have that u_x, u_y, v_x, v_y exist and are continuous. From real analysis, we know that this implies that u and v are totally differentiable.

Hence,

$$u(z_0 + h) - u(z_0) = \partial_x u(z_0)h_1 + \partial_y u(z_0)h_2 + |h|\epsilon_1(h),$$

where $\epsilon_1(h) \rightarrow 0$ as $h \rightarrow 0$. Similarly,

$$v(z_0 + h) - v(z_0) = \partial_x v(z_0)h_1 + \partial_y v(z_0)h_2 + |h|\epsilon_2(h).$$

Combining these two equations, we have, letting $\epsilon(h) = \epsilon_1(h) + i\epsilon_2(h)$,

$$\begin{aligned} f(z_0 + h) - f(z_0) &= (u + iv)(z_0 + h) - (u + iv)(z_0) \\ &= (\partial_x u + i\partial_x v)h_1 + (\partial_y u + i\partial_y v)h_2 + |h|\epsilon(h) \\ &= (\partial_x u - i\partial_y u)h_1 + (\partial_y u + i\partial_x u)h_2 + |h|\epsilon(h) \\ &= (\partial_x u - i\partial_y u)(h_1 + ih_2) + |h|\epsilon(h). \end{aligned}$$

Since $\epsilon(h) \rightarrow 0$ as $h \rightarrow 0$, we have that

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} = \partial_x u - i\partial_y u.$$

This implies, that f is differentiable at z_0 and its derivative is given by

$$\partial_x u - i\partial_y u = 2 \frac{\partial}{\partial z} u.$$

□

Example 1.15: Application of Cauchy-Riemann

Let $f(z) = \underbrace{x^2 + y^2}_u + i \underbrace{2xy}_v$. When is f holomorphic?

We have

$$u_x = 2x, \quad u_y = 2y, \quad v_x = 2y, \quad v_y = 2x.$$

So u, v are continuously differentiable. They satisfy the Cauchy-Riemann equations if and only if $2x = 2x$ and $2y = -2y$, which is true if and only if $y = 0$. Hence, f is holomorphic on the real line.

We've seen that for f holomorphic, $J_f = \begin{pmatrix} u_x & -v_x \\ v_x & u_x \end{pmatrix}$. Recall from linear algebra, that a linear map of the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ is a map which preserves angles, as it is a composition of a rotation and a scaling. This follows, since a linear map $z \mapsto (z)(a + ib)$, when written in polar coordinates, is a rotation by $\arg(a + ib)$ and a scaling by $|a + ib|$.

Hence, a holomorphic map, locally preserves angles.

1.6 Power Series

A power series is a series of the form

$$\sum_{n=0}^{\infty} a_n z^n.$$

Theorem 1.16:

Let $\sum a_n z^n$ be a power series. Then $\exists R \in \mathbb{R}_{\geq 0} \cup \{\infty\}$ such that

1. If $|z| < R$, then the series converges absolutely.
2. If $|z| > R$, then the series diverges.

Moreover, with the convention that $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$, R is given by

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

R is called the **RADIUS OF CONVERGENCE** of the power series. $D_R(0)$ is called the **DISK OF CONVERGENCE**.

Proof. The proof of this is the same as in real analysis, and is left as an exercise. □

Example 1.17: Exponential Function

The complex exponential function is defined by the power series

$$\exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

It converges absolutely for all $z \in \mathbb{C}$, since

$$\limsup_{n \rightarrow \infty} |1/n!|^{1/n} = 0.$$

We will see, that e^z in particular and power series in general give holomorphic functions in their disk of convergence.

Theorem 1.18:

The power series $\sum_{n=0}^{\infty} a_n z^n$ defines a holomorphic function on $D_R(0)$. Its derivative is given by

$$\left(\sum_{n=0}^{\infty} a_n z^n \right)' = \sum_{n=1}^{\infty} n a_n z^{n-1}.$$

Moreover, f' has the same disk of convergence as f .

Proof. Let R be the radius of convergence of f . Since $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$, we have that $\limsup_{n \rightarrow \infty} |n a_n|^{1/n} = \limsup_{n \rightarrow \infty} |a_n|^{1/n} = \frac{1}{R}$. Hence, the series for f' has the same radius of convergence as f . Let $z \in \mathbb{C}$, such that $|z| < R$. Choose δ such that $|z| + \delta < R$. Then, for $|h| < \delta$, we want to show that $\frac{f(z+h) - f(z)}{h} - \sum_{n=1}^{\infty} n a_n z^{n-1} \rightarrow 0$ as $h \rightarrow 0$. We have

$$\begin{aligned} & \left| \sum_{n=0}^{\infty} \frac{a_n(z+h)^n - a_n z^n}{h} - \sum_{n=1}^{\infty} n a_n z^{n-1} \right| \\ & \leq \sum_{n=0}^{\infty} |a_n| \left| \frac{1}{h} \left(\sum_{k=0}^n \binom{n}{k} z^{n-k} h^k - z^n \right) - n z^{n-1} \right| \\ & = \sum_{n=0}^{\infty} |a_n| \left| \sum_{k=2}^n \binom{n}{k} h^{k-1} z^{n-k} \right|. \end{aligned}$$

Since the first two terms cancel. Now, pulling out the first two terms of the binomial coefficient, we have

$$\begin{aligned} & \sum_{n=0}^{\infty} |a_n| \left| \sum_{k=2}^n \binom{n}{k} h^{k-1} z^{n-k} \right| \\ & \leq \sum_{n=0}^{\infty} |a_n| (n(n-1)) \sum_{k=2}^n \binom{n-2}{k-2} |h|^{k-2} |z|^{n-k} |h| \\ & \leq |h| \sum_{n=0}^{\infty} |a_n| n(n-1) (|z| + |h|)^{n-2} \\ & \leq |h| \sum_{n=0}^{\infty} |a_n| n(n-1) (|z| + \delta)^{n-2} \end{aligned}$$

Since $|z| + \delta < R$, the series $\sum_{n=0}^{\infty} |a_n| n(n-1) (|z| + \delta)^{n-2}$ converges. Hence, we have that this expression goes to 0 as $h \rightarrow 0$. Hence, f is holomorphic with the derivative given by the termwise derivative of the power series. $\square$

Example 1.19:

The exponential function is an entire function and its derivative is given by

$$\frac{d}{dz} e^z = \sum_{n=1}^{\infty} \frac{z^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{z^{n-1}}{(n-1)!} = \sum_{m=0}^{\infty} \frac{z^m}{m!} = e^z.$$

Example 1.20: Trigonometric Functions

We define the complex sine and cosine functions as

$$\begin{aligned} \cos(z) &:= \frac{e^{iz} + e^{-iz}}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}, \\ \sin(z) &:= \frac{e^{iz} - e^{-iz}}{2i} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}. \end{aligned}$$

Both of these functions are entire.

As a warning, the complex sine and cosine functions are not bounded, unlike their real counterparts!

Example 1.21: Geometric Series

The geometric series is given by

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z},$$

which converges for $|z| < 1$.

1.7 Complex Line Integrals

We like to recap some definitions.

Definition 1.22: Parametrized Curve

A **PARAMETRIZED CURVE** in $\mathbb{C}$ is a continuous function $\gamma : [a, b] \rightarrow \mathbb{C}$.

We call a curve **SMOOTH** if its derivative $\gamma'(t)$ exists and $\forall t \in [a, b]$, $\gamma'(t)$ is continuous and $\gamma'(t) \neq 0$.

A **PIECEWISE SMOOTH** curve is a curve γ such that $\exists$ points $a = a_0 < a_1 < \dots < a_n = b$ such that $\gamma|_{[a_i, a_{i+1}]}$ is smooth for all $i = 0, \dots, n-1$.

A **CLOSED CURVE** is a curve γ such that $\gamma(a) = \gamma(b)$. A curve is **SIMPLE** if it is not self-intersecting.

In the following we will always assume that a curve is piecewise smooth.

Definition 1.23: Reparametrization

A **REPARAMETRIZATION** of a curve $\gamma : [a, b] \rightarrow \mathbb{C}$ is a curve

$$\tilde{\gamma} : [c, d] \rightarrow \mathbb{C},$$

such that there exists a continuously differentiable function $\sigma : [c, d] \rightarrow [a, b]$ which is bijective and $\sigma'(t) > 0$ for all $t \in [c, d]$ and $\tilde{\gamma}(t) = \gamma(\sigma(t))$.

Example 1.24: Parametrized Line Segment

Given 2 points $z_1, z_2 \in \mathbb{C}$, the path

$$\gamma : [0, 1] \rightarrow \mathbb{C}, t \mapsto \gamma(t) = (1-t)z_1 + tz_2,$$

is the line segment between z_1 and z_2 .

Example 1.25: Parametrized Circle

A circle of center $z_0 \in \mathbb{C}$ and radius $r > 0$ can be parametrized by

$$\gamma : [0, 2\pi] \rightarrow \mathbb{C}, t \mapsto \gamma(t) = z_0 + re^{it}.$$

We want to define path integrals of complex functions. Recall that if g is a real-valued continuous function of a real variable, then it is Riemann-integrable. If g is complex valued $g : [a, b] \rightarrow \mathbb{C}, g(t) = u(t) + iv(t)$, then we define

$$\int_a^b g(t) dt = \int_a^b u(t) dt + i \int_a^b v(t) dt.$$

Definition 1.26: Line Integral

Suppose $\gamma : [a, b] \rightarrow \Omega \subset \mathbb{C}$ is a piecewise smooth curve and $f : \Omega \rightarrow \mathbb{C}$ is defined and continuous on the curve. Then the **LINE INTEGRAL** of f along γ is defined by

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

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